

24 – M4 Trading Documents

Module design for the inter-entity flow from order to settlement: order, delivery note with drops, goods received note as the ownership pivot, discrepancy and claim, invoice, credit and debit notes, payment receipt with settlement to specific invoices, statements, and the warn-only exposure picture. This is where the state machines the Stage 2 plan called for live; numbering detail waits for the tax advisor (24B).

Field	Value
Document	24 – M4 Trading Documents (Stage 2 module design; template of doc 20 §5)
Version	v0.1 – draft for design review; contract freeze targeted end of Stage 2 week 5
Date	9 September 2026
Owning developer	Module developer (full-stack) for the back office and the driver app screens; the till track hosts the GRN screens in shop-management mode from this module's specification
Inputs	05 SRS (FR-ORD, FR-DEL, FR-BIL); 03 state machines §2–8; 13 scenarios 1 and 2; 16 answers (A, C, D, I-05 to I-07, J sections); 09 ADR-01, 03, 04, 11, 12; 17 v0.2.1 §7.4, ADR-37; 18 v0.2 part C; 19 v0.2; 20 v0.3 M4 blueprint; 21–23 contracts; 30 §5.4; 32
Depends on	M1 (relationships and terms, locations, scope), M2 (SKU, conversions, batch registration), M3 (trade price, adjustment types), M5 query (availability), kernel (document base, numbering, links, attachments, audit, events)
Depended on by	M5 (stock effects of GRN, returns), M2 (batch registration), M7 (customer payments share the PRC base), M8, M9 (journal export, e-invoice seam, notifications), M10 (purchase-order receipts, phase 2)

One event carries the whole module. `grn.confirmed.v1` is the moment ownership and risk transfer (firm sale, ADR-03). Before it, the goods are the seller's problem; after it, the buyer's. Batches get registered from it, lots get created from it, invoicing is permitted by it, and loss attribution depends on its timestamp. Everything in this module is either on the way to that event or a consequence of it.

1. Purpose and scope

M4 owns the documents that pass between two legal entities and the ledger picture they produce: what was ordered, what was dispatched, what was received, what is disputed, what is owed and what has been paid. It never moves money (ADR-01) and never blocks on credit (ADR-12); it records, links and computes.

In scope (phase 1)	Out of scope
Order (buyer's document): manual lines and system suggestions (P2), submission, seller acceptance with allocation (FCFS default, manual override), partial fulfilment with backorders, indicative ETA, amend/cancel until the lock window, manual adjustments on Federation orders (C-01 to C-09, A-06)	Vehicle routing, load planning, fleet management (D-01); minimum order quantities (C-06); fixed ordering cycles (C-07)
Delivery note with drops (ship-to per drop, bill-to entity), vehicle, driver, timestamps; proof of delivery by driver app or paper (D-02, D-03, D-09)	Returnable containers (D-10); cold-chain telemetry
GRN: actuals recorded by the receiver, never forced to match; offline capture and confirmation at a shop; the ownership pivot; reversal within a window (D-04, D-05, ADR-03)	Stock lots and movements themselves (M5 applies them from the event)
Discrepancy raised automatically on any mismatch; resolution by the parties within the window; escalation to the Federation as arbiter (D-05, D-06)	–
Claim with photographs: raise, inspect, approve or reject; credit note and return on approval (F-03, D-07, D-08)	Insurance or transporter recovery outside the system
Invoice following GRN; credit and debit notes at every tier; tax invoice with both VAT numbers; immutable snapshot for e-invoicing readiness (J-03, J-04, J-07)	Statutory accounting (ADR-05); interest posting automation (A-12 deferred); live e-invoicing (P3)
Payment receipt recorded by the seller, settled to specific invoices; cheque sub-lifecycle; dispute and reversal (I-05, I-07)	Bank integration or payment processing (ADR-01); customer payments' screens (M7 reuses the base)
Statements per relationship at a configured frequency; committed exposure warn-only with freshness (I-06, A-10)	Credit blocking of any kind
Journal-posting map per document type as data (J-02); lateral transfer request mediated by the MPCS (C-10, C-11)	The transfer document itself (M5 XFR)
Numbering: uses the kernel series (buyer's ENTITY series for orders, seller's for DN/INV/CN/DN2/PRC, receiver's LOCATION series for shop GRNs)	Numbering rules pending the tax advisor: addendum 24B

2. Requirements coverage

Identifier	Requirement (abridged)	Design section
FR-ORD-010	Requisition by manual entry and system suggestion (C-01)	3.1, 6.1
FR-ORD-020	Seller approves manually; multi-role approval; no auto-fulfilment (C-02)	4.1
FR-ORD-030	Allocation FCFS default, manual override, configurable per seller, audited (C-03, C-04)	3.1, 6.2
FR-ORD-040	Partial fulfilment; remainder stays open as backorder unless cancelled (C-05)	3.1, 4.1
FR-ORD-050	Amend/cancel until a configurable lock window before ETA; per-order override (C-08)	4.1, 7
FR-ORD-060	Indicative ETA at acceptance; delivery performance measured (C-13)	3.1, 5.2
FR-ORD-070	Buyer sees seller availability, live or indicative (C-12)	6.1
FR-ORD-080	Lateral transfer request within an MPCS, mediated (C-10, C-11)	3.7, 4.7
FR-DEL-010	Delivery note per dispatch; multi-drop from one vehicle (D-02)	3.2
FR-DEL-020	Bill-to and ship-to distinguished on drops	3.2
FR-DEL-030	Receiver records actuals; never forced to match (D-04, D-05)	3.3
FR-DEL-040	Discrepancy raised automatically on mismatch (D-05)	3.4, 4.4
FR-DEL-050	Photographs attachable to discrepancies and claims (D-08)	3.4, 3.5
FR-DEL-060	GRN capture and confirmation offline (H-02)	3.3, 9.2
FR-BIL-010	Invoice from received goods at the relationship's trade price	3.6
FR-BIL-020	Journal-posting map per document type (J-02)	3.9
FR-BIL-030	Credit and debit notes at all tiers (J-07)	3.6, 4.6

Identifier	Requirement (abridged)	Design section
FR-BIL-040	Statements per relationship (I-06)	3.8
FR-BIL-050/060	Series scopes and gapless numbering (ADR-11)	3.10, 24B
FR-BIL-070	Payment matched to specific invoices (I-07)	3.7, 4.6
FR-BIL-080	Exposure computed and warned, never blocked (ADR-12)	3.8
FR-BIL-090	Counter transfer on device replacement	kernel (doc 32 §8)
FR-BIL-100	Interest rule configurable, manual application (A-12)	7
FR-BIL-110	E-invoicing readiness: immutable invoice snapshot (J-04)	3.6, 9.5
FR-BIL-120/130	Tax invoice content; VAT numbers of both parties (J-03)	3.6
FR-BIL-140	Postings derived from documents	3.9
FR-BIL-150	Cheque sub-lifecycle	4.6
ADR-03	Ownership and risk at GRN confirmation	3.3
D-06	Discrepancy time limit; Federation arbiter	4.4, 7

3. Domain model

All documents use the kernel base (doc 18 part C): header, lines, typed links, state history, attachments. M4 adds extension tables in schema trading, keyed one-to-one on document_id and inheriting immutability.

3.1 Order (ORD)

Aspect	Design
Extension	doc_order(document_id, relationship_id, requested_eta, committed_eta, lock_at, lock_override_by, allocation_run_id, backorder_of_order_id); line extension doc_order_line(line_id, requested_qty, allocated_qty, fulfilled_qty, backorder_qty, tier_price, adjustment_type?, adjustment_value?).
Issuer and series	The buyer issues; ENTITY series of the buyer (<i>doc 18 §5.5 corrected</i>).
Allocation run	allocation_run(seller, at, rule, orders considered, availability snapshot, manual overrides): FCFS by submission time by default; a seller may override allocations with a reason; the run is audited (C-04).
Backorders	Unallocated remainder stays open on the same order as backorder_qty until the seller allocates it in a later run or either party cancels it; a configurable age (default 30 days) raises a REVIEW, never an auto-cancel.
Lock	lock_at = committed_eta – relationship.order_lock_hours_before_eta; a permitted seller role may override per order with a reason (C-08).
Adjustments	Only on Federation→distributor orders, against M3 adjustment types and within their bounds, gated by the type's permission (A-06); recorded as a line of kind ADJUSTMENT that flows to the invoice.

3.2 Delivery note (DN)

Aspect	Design
Extension	doc_delivery(document_id, vehicle_ref, driver_user_id?, driver_name, dispatched_at, route_ref); doc_delivery_drop(drop_id, document_id, seq, ship_to_location_id, bill_to_entity_id, order_ids[], status, arrived_at, pod_kind PAPER APP, pod_signature_key?, pod_photo_key?, delivered_by); lines carry drop_id, dispatched_qty, batch_id.
Direct drop	One DN from one vehicle may serve several drops; a drop's ship-to may be a shop location of the bill-to entity (direct-to-shop, D-02). The GRN for that drop is captured at the shop, in the shop's LOCATION series, offline-capable.
Risk	Remains with the seller until each drop's GRN is confirmed (A-03); in-transit loss is the seller's.

3.3 Goods received note (GRN)

Aspect	Design
Extension	doc_grn(document_id, drop_id?, delivery_document_id?, purchase_order_ref?, received_at, counted_by, confirmed_by, confirmed_at, reversal_of?); lines carry expected_qty (from the DN drop, nullable for local supply), received_qty, batch_id (registered through M2 at confirmation), condition GOOD DAMAGED, damaged_qty.
The pivot	Confirmation is the transfer of ownership and risk (ADR-03). In the same transaction: batches registered (M2), grn.confirmed.v1 published; M5 creates the receiving entity's lots at the trade price and posts receipt movements; discrepancy raised if any line's received ≠ expected or any damaged_qty > 0.
Offline	Captured and confirmed on the till in shop-management mode against the DN drop in the snapshot (or without expectation for local supply); numbered from the shop's LOCATION series held by the primary till; ingested through the document bundle path (doc 32 §3.1); the confirmation timestamp is the till's, corrected for clock offset.
Reversal	A confirmed GRN may be reversed within grn_reversal_hours (default 24) only if no sale has consumed its lots (M5 query) and no invoice references it; reversal is a REVERSES-linked GRN with negated lines; ownership reverts. Beyond the window, corrections go through discrepancy or claim.
Local supply	An MPCS receiving open-market goods (A-08) issues a GRN with no DN, a supplier from M2's minimal table, and its own cost per line; the same pivot applies (the entity now owns the goods).

3.4 Discrepancy

Aspect	Design
Document	<code>doc_discrepancy(document_id, grn_document_id, delivery_document_id, raised_at, window_ends_at, kind SHORT OVER DAMAGED MIXED, resolution, resolved_by, escalated_at, arbiter_user_id)</code> with lines (sku, batch, expected, received, damaged, variance); DISPUTES-linked to the GRN and the DN (<i>doc 18 document link</i>).
Automatic	Raised by the GRN confirmation handler; no user action; the seller is notified (R-08).
Resolution	Within <code>relationship.discrepancy_window_days</code> : ACCEPT_SHORT (seller issues a credit note for the shortfall), ACCEPT_OVER (seller invoices the extra or buyer returns), REJECT (seller disputes the count; buyer may raise a claim), SPLIT (agreed quantities). After the window without resolution: escalation to the Federation arbiter (D-06), whose decision is recorded and binding.

3.5 Claim

Aspect	Design
Document	<code>doc_claim(document_id, grn_document_id, discrepancy_document_id?, kind DAMAGED EXPIRED_ON_ARRIVAL WRONG_GOODS QUALITY, raised_at, inspected_at, inspector, decision, decided_at, return_required)</code> with lines (sku, batch, qty, value at trade price) and attachments (photographs mandatory, F-05 style).
Outcome	APPROVED → seller issues a credit note (CREDITS link) and, if <code>return_required</code> , M5 posts a return movement on <code>claim.approved.v1</code> ; REJECTED → closed with reason; either may be escalated like a discrepancy.

3.6 Invoice, credit note, debit note

Aspect	Design
Invoice	<code>doc_invoice(document_id, grn_document_ids[], delivery_document_id?, seller_vat_no, buyer_vat_no, tax_point_date, due_date, settled_amount, e_invoice_status)</code> ; lines from the GRN's received quantities at the trade price resolved through M3 (tier by the ordered quantity, DR-2) plus adjustment lines; VAT per line from the SKU's category; immutable snapshot (<code>content_hash</code> , header and lines) is the e-invoicing readiness (J-04).
Timing	Phase 1: invoice follows GRN (one invoice per GRN or a consolidated invoice per DN across its drops for one bill-to, seller's choice). Invoice-on-dispatch is deferred (doc 20 DR-2).
Credit note	Issued by the seller against an invoice (CREDITS link with amount) for discrepancy shortfalls, approved claims, price corrections; reduces the invoice's open balance; an unapplied credit note (issued before the invoice) is applied later with a link.
Debit note	Issued by the seller for under-billing or agreed charges (interest applied manually, A-12), DEBITS-linked to the original invoice.

3.7 Payment receipt and cheque

Aspect	Design
Document	<code>doc_payment_receipt(document_id, relationship_id, payer_entity_id, method CASH CHEQUE TRANSFER DEPOSIT, reference, received_on, amount, unapplied_amount)</code> ; SETTLES links carry the amount applied per invoice (I-07); never oldest-first; unapplied amounts sit on the receipt until applied.
Cheque	<code>cheque(receipt_document_id, bank, number, dated, status RECEIVED CLEARED BOUNCED, outcome_at)</code> ; a bounce reverses the settlements (REVERSES link on the receipt) and raises an ALERT; the invoices reopen.
Recorded only	The seller records what was received; the system processes nothing (ADR-01). M7 reuses this base for customer payments with the customer account as payer.

3.8 Exposure and statements

Aspect	Design
Exposure	<code>committed_exposure(relationship) = open invoice balances + accepted-not-invoiced order value – unapplied credit notes – unapplied receipts</code> ; recomputed on every relevant event into <code>relationship_exposure(relationship_id, amount, as_of, freshness)</code> ; compared with <code>relationship.credit_limit</code> to raise <code>exposure.warning.v1</code> (warn only, ADR-12) at configurable thresholds (80%, 100%).

Aspect	Design
Statement	Generated per relationship for a period from documents and links: opening balance, invoices, credit and debit notes, receipts with allocations, closing balance and ageing buckets; rendered A4 (kernel) and notified (R-08); a report, not a document with a number.

3.9 Journal-posting map

Aspect	Design
Data	<code>posting_map(doc_type, line_kind, debit_account_role, credit_account_role, amount_source)</code> seeded per document type (INV: receivable/revenue/VAT output; CN: reverse; GRN at the buyer: inventory/payable; PRC: bank/receivable) using account <i>roles</i> , mapped to a chart of accounts when the accounting system is chosen (J-02). M9 exports <code>journal.postings_ready.v1</code> payloads.

3.10 Numbering (interim)

Aspect	Design
Series	ORD: buyer ENTITY; DN, INV, CN, DN2, PRC: seller ENTITY; GRN: receiving LOCATION when captured at a shop, receiving ENTITY at a warehouse; discrepancy and claim: issuer ENTITY (<i>doc 18 §5.5</i>). Financial-year reset and fiscal scope await the tax advisor (24B); until then no reset, GRN non-fiscal.

4. State machines

Reconciled with doc 03 sections 2–8. Every transition has a guard, a permission and an audit event; offline-issuable transitions are marked (H-02).

4.1 Order

From	To	Trigger / command	Guard	Permission	Audit / offline
–	DRAFT	CreateOrder	buyer scope; ACTIVE relationship (M1); lines with active SKUs	ord.order.draft	–
DRAFT	SUBMITTED	SubmitOrder	≥1 line; qty > 0; unit valid (M2 conversion); issuance protocol (buyer series)	ord.order.submit	ORDER_SUBMITTED / online only
SUBMITTED	ACCEPTED	AcceptOrder(allocations, eta)	seller scope; allocation run recorded; eta ≥ today; lock_at computed	ord.order.accept	ORDER_ACCEPTED
SUBMITTED	REJECTED	RejectOrder(reason)	seller; reason	ord.order.accept	ORDER_REJECTED
SUBMITTED / ACCEPTED	AMENDED (new version)	AmendOrder	buyer; before lock_at (or override); an amendment SUPERSEDES the previous lines (draft-level supersession, number retained)	ord.order.submit; ord.order.lock_override (seller)	ORDER_AMENDED
ACCEPTED	LOCKED	Clock (lock_at)	–	system	ORDER_LOCKED
ACCEPTED / LOCKED	PARTIALLY_FULFILLED	DN issued for part of the allocation	–	system (from DN)	–
PARTIALLY_FULFILLED / LOCKED	FULFILLED	All allocated qty on issued DNs; backorder_qty = 0 or cancelled	–	system	ORDER_FULFILLED
ACCEPTED (with backorder)	ACCEPTED	AllocateBackorder (later run)	seller; availability	ord.order.accept	ORDER_ALLOCATED
any non-terminal	CANCELLED	CancelOrder(reason)	buyer before lock; seller any time with reason; only unallocated/undispatched quantity cancels (partial cancel of the remainder)	ord.order.submit / ord.order.accept	ORDER_CANCELLED

4.2 Delivery note

From	To	Trigger / command	Guard	Permission	Audit / offline
–	DRAFT	CreateDeliveryNote(orders, drops)	seller; orders ACCEPTED/LOCKED with allocations; batches available (M5); drops' ship-to belong to bill-to	del.note.draft	–
DRAFT	ISSUED	IssueDeliveryNote	issuance (seller series); M5 reserves/picks the batches (pick list)	del.note.issue	DN_ISSUED
ISSUED	IN_TRANSIT	Dispatch(vehicle, driver, at)	–	del.note.dispatch	DN_DISPATCHED
IN_TRANSIT	IN_TRANSIT (drop delivered)	RecordProofOfDelivery(drop)	driver app or paper; signature/photo optional	del.pod.record (driver)	DROP_DELIVERED / offline (driver app)
IN_TRANSIT	CLOSED	All drops have a confirmed GRN, or a drop is marked UNDELIVERED with reason (goods returned to seller: M5 return-to-stock)	–	system / del.note.dispatch	DN_CLOSED

4.3 Goods received note

From	To	Trigger / command	Guard	Permission	Audit / offline
–	DRAFT	CaptureGrn(drop local supply)	receiver scope; expected lines from the drop when present	shop.grn.confirm (shop) / whs.grn.confirm (warehouse)	– / offline
DRAFT	CONFIRMED	ConfirmGrn	received_qty per line; batch data per line (MRP when has_printed_mrp, expiry when tracked); issuance (LOCATION or ENTITY series); the pivot (Section 3.3)	as above	GRN_CONFIRMED / offline
CONFIRMED	DISCREPANCY_RAISED	system (on confirm) when any variance or damage	–	system	DISCREPANCY_RAISED
CONFIRMED	CLOSED	system when no variance	–	system	GRN_CLOSED
DISCREPANCY_RAISED	CLOSED	Discrepancy resolved or escalation decided	–	system	GRN_CLOSED
CONFIRMED / DISCREPANCY_RAISED	REVERSED	ReverseGrn(reason)	within grn_reversal_hours; no lot consumed (M5); no invoice references it; REVERSES-linked negated GRN issued	shop.grn.reverse (supervisor, online)	GRN_REVERSED / online only

4.4 Discrepancy

From	To	Trigger / command	Guard	Permission	Audit / offline
–	RAISED	system (GRN confirm)	variance or damage present; window_ends_at = confirm + discrepancy_window_days	system	DISCREPANCY_RAISED
RAISED	UNDER_REVIEW	AcknowledgeDiscrepancy	seller scope	del.discrepancy.resolve (seller)	–
RAISED / UNDER_REVIEW	RESOLVED	ResolveDiscrepancy(resolution, agreed lines)	either party proposes, the other accepts (two-step: PROPOSED then ACCEPTED); ACCEPT_SHORT triggers seller credit note; ACCEPT_OVER triggers extra invoice line or return	del.discrepancy.resolve (both)	DISCREPANCY_RESOLVED
RAISED / UNDER_REVIEW	ESCALATED	Clock (window_ends_at) or EscalateDiscrepancy	window passed or either party escalates with reason	system / del.discrepancy.escalate	DISCREPANCY_ESCALATED
ESCALATED	RESOLVED	ArbitrateDiscrepancy(decision, lines)	Federation arbiter role; decision binding; reason	fed.arbitrate (F, MFA)	DISCREPANCY_ARBITRATED
RESOLVED	CLOSED	Consequent documents issued (credit note, return, extra invoice) or none required	–	system	DISCREPANCY_CLOSED

4.5 Claim

From	To	Trigger / command	Guard	Permission	Audit / offline
–	RAISED	RaiseClaim(grn, lines, kind, photos)	buyer scope; within claim window (config); photographs attached (pending upload allowed, must be COMPLETE before decision)	del.claim.raise	CLAIM_RAISED / draft offline, issue online
RAISED	INSPECTED	InspectClaim(findings)	seller scope; attachments COMPLETE	del.claim.inspect (seller)	CLAIM_INSPECTED
INSPECTED	APPROVED	ApproveClaim(lines, return_required)	seller; credit note issued in the same transaction (CREDITS link)	del.claim.decide (seller)	CLAIM_APPROVED
INSPECTED	REJECTED	RejectClaim(reason)	seller; reason	del.claim.decide	CLAIM_REJECTED
RAISED / INSPECTED / REJECTED	ESCALATED	EscalateClaim	either party; reason	del.claim.escalate	CLAIM_ESCALATED

From	To	Trigger / command	Guard	Permission	Audit / offline
ESCALATED	APPROVED / REJECTED	ArbitrateClaim	Federation arbiter	fed.arbitrate (F, MFA)	CLAIM_ARBITRATED
APPROVED / REJECTED	CLOSED	Return movement posted (M5) if required; else immediate	–	system	CLAIM_CLOSED

4.6 Invoice, credit note, payment receipt

From	To	Trigger / command	Guard	Permission	Audit / offline
–	INV ISSUED	IssueInvoice(grns dn)	seller; GRNs CONFIRMED or CLOSED; not already invoiced; trade prices resolved (M3); VAT numbers present; issuance (seller series); snapshot hashed	bil.invoice.issue	INVOICE_ISSUED / online only
INV ISSUED	PARTIALLY_SETTLED / SETTLED	SETTLES links from receipts or CREDITS links from credit notes	sum of applied ≤ invoice total	system	–
INV ISSUED / PARTIALLY_SETTLED	DISPUTED	DisputeInvoice(reason)	buyer scope; reason	bil.invoice.dispute (buyer)	INVOICE_DISPUTED
DISPUTED	ISSUED / PARTIALLY_SETTLED	ResolveInvoiceDispute	seller with buyer acceptance, or arbiter	bil.invoice.issue / fed.arbitrate	INVOICE_DISPUTE_RESOLVED
–	CN / DN2 ISSUED	IssueCreditNote / IssueDebitNote(invoice?, lines, reason)	seller; MFA (credit note); amount ≤ open balance when linked; reason	bil.creditnote.issue (MFA) / bil.debitnote.issue	CREDIT_NOTE_ISSUED / DEBIT_NOTE_ISSUED
CN ISSUED	APPLIED	ApplyCreditNote(invoice, amount)	open balance	bil.creditnote.issue	CREDIT_NOTE_APPLIED
–	PRC RECORDED	RecordPaymentReceipt(method, amount, settlements[])	seller; each settlement ≤ invoice open balance; cheque details when CHEQUE; issuance (seller series)	bil.payment.record	PAYMENT_RECORDED
PRC RECORDED	DISPUTED	DisputePayment(reason)	either party	bil.payment.record / bil.invoice.dispute	PAYMENT_DISPUTED
PRC RECORDED / DISPUTED	REVERSED	ReversePaymentReceipt(reason) or cheque BOUNCED	seller; MFA; REVERSES link; settlements undone; invoices reopen	bil.payment.reverse (MFA)	PAYMENT_REVERSED
Cheque RECEIVED	CLEARED / BOUNCED	RecordChequeOutcome	seller; bounce triggers reversal	bil.payment.record	CHEQUE_CLEARED / CHEQUE_BOUNCED

4.7 Lateral transfer request (within an MPCs)

From	To	Trigger / command	Guard	Permission	Audit / offline
–	REQUESTED	RequestLateralTransfer(from shop, to shop, lines)	requesting shop scope; both shops of one MPCs	shop.transfer.request	TRANSFER_REQUESTED / online
REQUESTED	APPROVED	ApproveLateralTransfer	MPCs scope; availability at source (M5)	mpcs.transfer.approve	TRANSFER_APPROVED (M5 issues XFR at cost)
REQUESTED	REJECTED	RejectLateralTransfer(reason)	MPCs	mpcs.transfer.approve	TRANSFER_REJECTED

5. Contract

5.1 Commands

Command	Input (essentials)	Permission	Events
CreateOrder / SubmitOrder / AmendOrder / CancelOrder	relationship, lines (sku, uom, qty), requested_eta; reason	ord.order.draft / submit	order.submitted/amended/cancelled.v1
AcceptOrder / RejectOrder / AllocateBackorder	order, allocations per line, committed_eta, overrides with reasons	ord.order.accept	order.accepted/rejected/allocated.v1
OverrideLockWindow	order, new lock_at, reason	ord.order.lock_override	order.lock_overridden.v1
RecordOrderAdjustment	order, adjustment type, value	type's permission (M3)	order.adjusted.v1
CreateDeliveryNote / IssueDeliveryNote / Dispatch	orders, drops (ship-to, bill-to), lines with batches; vehicle, driver	del.note.*	delivery_note.issued/dispatched.v1
RecordProofOfDelivery	drop, kind, signature/photo keys, at, location	del.pod.record	drop.delivered.v1
MarkDropUndelivered	drop, reason	del.note.dispatch	drop.undelivered.v1
CaptureGrn / ConfirmGrn	drop or local supply; lines (received, batch data, condition)	shop.grn.confirm / whs.grn.confirm	grn.captured/confirmed.v1 (+ discrepancy.raised.v1)
ReverseGrn	grn, reason	shop.grn.reverse	grn.reversed.v1
AcknowledgeDiscrepancy / ProposeResolution / AcceptResolution / EscalateDiscrepancy / ArbitrateDiscrepancy	discrepancy, resolution kind, agreed lines, reason	del.discrepancy.* / fed.arbitrate	discrepancy.*.v1
RaiseClaim / InspectClaim / ApproveClaim / RejectClaim / EscalateClaim / ArbitrateClaim	grn, lines, kind, photos; findings; decision	del.claim.* / fed.arbitrate	claim.*.v1
IssueInvoice / DisputeInvoice / ResolveInvoiceDispute	grns or dn, consolidation choice; reason	bil.invoice.*	invoice.issued/disputed/resolved.v1
IssueCreditNote / ApplyCreditNote / IssueDebitNote	invoice?, lines or amount, reason	bil.creditnote.issue (MFA) / bil.debitnote.issue	credit_note.issued/applied.v1; debit_note.issued.v1
RecordPaymentReceipt / ApplyReceipt / DisputePayment / ReversePaymentReceipt / RecordChequeOutcome	method, amount, reference, settlements; cheque details; reason	bil.payment.*	payment_receipt.recorded/applied/disputed/reversed.v1; cheque.cleared/bounced.v1
GenerateStatement	relationship, period	bil.statement.generate	statement.generated.v1
RequestLateralTransfer / ApproveLateralTransfer / RejectLateralTransfer	shops, lines; reason	shop.transfer.request / mpcs.transfer.approve	transfer_request.*.v1

5.2 Queries

Query	Returns	Scope rule
GetOrder / ListOrders(filter)	Header, lines, allocations, status, ETA, lock	PARTY (buyer sees its orders; seller sees orders to it)
SellerAvailability(seller, skus)	Live or indicative on-hand (seller's choice in configuration) via M5	Buyer with an ACTIVE relationship
ListOpenBackorders(seller) / ListDueOrders(seller, date)	For allocation runs and DN creation	OWN
GetDeliveryRun(vehicle, date) / GetDrop(drop)	DN with drops for the driver app; expected lines for GRN capture (snapshot)	OWN; driver scope; receiver scope for its drop
GetGrn / ListGrns(location, period)	GRNs with lines and discrepancy status	OWN, PARTY (seller sees GRNs against its DNs)
ListDiscrepancies / ListClaims(scope, status)	Open items with windows and escalation state	PARTY; FEDERATION_VIEW; arbiter
GetInvoice / ListInvoices(relationship, status)	Header, lines, links, open balance	PARTY (masked: no cost columns)
ListUnsettledInvoices(relationship)	For payment matching	PARTY

Query	Returns	Scope rule
GetExposure(relationship) / ListExposures(seller)	Amount, components, limit, freshness	OWN (seller), FEDERATION_VIEW; buyer sees its own
GetStatement(relationship, period)	Rendered statement reference and data	PARTY
DeliveryPerformance(seller, period)	Committed vs actual per order (C-13)	OWN, FEDERATION_VIEW
PostingsFor(document)	Journal lines by account role (M9)	M9 internal

5.3 Events published

Event	Payload (beyond envelope)	Consumers
order.submitted/accepted/rejected/amended/allocated/locked/cancelled/adjusted.v1	order id, relationship, parties, lines summary, eta, lock	M5 (allocation availability), M8, notifications
delivery_note.issued/dispatched/closed.v1; drop.delivered/undelivered.v1	dn id, drops, ship-to, bill-to, batches	M5 (pick and return), snapshot (expected lines for GRN capture at the shop), M8, notifications
grn.captured/confirmed/closed/reversed.v1	grn id, receiver entity and location, seller, drop, lines with batch data and condition, confirmed_at	M2 (batch registration), M5 (lots, receipts, reversal), M3 (via M2 batch event), M8, M10
discrepancy.raised/acknowledged/proposed/resolved/escalated/arbitrated/closed.v1	ids, kind, lines, resolution	Notifications, exception queues, M8, arbiter queue
claim.raised/inspected/approved/rejected/escalated/arbitrated/closed.v1	ids, kind, lines, decision, return_required	M5 (return movement), notifications, M8
invoice.issued/disputed/resolved.v1	invoice id, parties, VAT numbers, totals, grns, snapshot hash	M9 (e-invoice seam), M8, notifications, exposure
credit_note.issued/applied.v1; debit_note.issued.v1	ids, invoice, amount, reason	Exposure, M8, M9
payment_receipt.recorded/applied/disputed/reversed.v1; cheque.cleared/bounced.v1	receipt id, payer, amount, settlements	Exposure, M7 (customer payer), M8, M9
exposure.updated/warning.v1	relationship, amount, limit, threshold, freshness	Order desk screen, notifications, M8
statement.generated.v1	relationship, period, object key	Notifications, M8
journal.postings_ready.v1	document id, type, postings by account role	M9
transfer_request.requested/approved/rejected.v1	shops, lines	M5 (XFR issuance on approval), notifications

6. Key flows

Flow	Sequence	Failure paths
6.1 Order to cash (doc 13 scenario 1)	MPCS stores role drafts an order from suggestions and manual lines → sees distributor availability (indicative) → submits (buyer series) → distributor runs allocation (FCFS) with a manual override for a scarce SKU (reason) → accepts 80% with ETA; 20% backorder → exposure recomputed and warned at 100% → DN issued with one drop to the MPCS warehouse → dispatched → GRN confirmed at the warehouse (ENTITY series) → invoice issued from received quantities at tiered trade price → statement at month end → cheque received and matched to the invoice → cleared	Order to a SUSPENDED relationship refused; allocation over availability refused; invoice before GRN refused; settlement over open balance refused; bounce reverses settlement and reopens the invoice
6.2 Direct drop and offline GRN (doc 13 scenario 2)	Distributor DN with three drops to three shops of one MPCS (bill-to MPCS) → driver app records POD at each drop offline → shop 2 captures the GRN on the till during an outage from the drop lines in its snapshot: counts actuals, keys batch MRP and expiry for a new batch, marks two damaged → confirms (LOCATION series, primary till) → on sync: bundle ingested, batches registered, lots created, discrepancy raised automatically for the damage → distributor acknowledges, proposes ACCEPT with credit for the damaged units → MPCS accepts → credit note CREDITS the invoice	GRN line without MRP for a has_printed_mrp SKU blocked on the till with the reason; a drop without snapshot data (DN issued after the last sync) is captured as local supply against the DN number keyed by hand and reconciled at ingestion (REVIEW)
6.3 Claim with photographs	MPCS raises a claim for expired-on-arrival goods within the window with photos (uploaded via presign; pending allowed) → distributor inspects once attachments are COMPLETE → approves with return required → credit note issued; M5 posts the return on claim.approved.v1	Decision refused while any attachment is PENDING; claim after the window refused with the window shown
6.4 Escalation to the Federation	Discrepancy unresolved after the window → system escalates; arbiter queue shows both parties' statements and evidence → arbiter decides (MFA) → consequent documents issued by the seller; refusal to issue is itself an exception item	–
6.5 Federation order with adjustment	Distributor orders from the uniform list → Federation sales applies SPECIAL_DISC 3% within bounds (permission) → invoice carries the adjustment line	Beyond bounds refused using M3's type query
6.6 GRN reversal	Warehouse confirmed a GRN against the wrong DN → supervisor reverses within 24 h (no lots consumed, no invoice) → REVERSES-linked GRN; ownership reverts; batches remain registered (identity is global)	After a sale from the lot: refused; correct via discrepancy or claim
6.7 Lateral transfer	Shop A requests 20 units from shop B → MPCS approves against B's availability → M5 issues XFR at cost; no invoice (C-11)	Cross-MPCS request refused: that is a sale
6.8 Statement and exposure	Month end job generates statements per relationship; exposure warning at 80% notifies the seller's accounts role; nothing blocks	Stale freshness (a shop not synced) is shown on the exposure, not hidden

7. Configuration

Item	Default	Scope	Who may change
relationship.discrepancy_window_days / order_lock_hours_before_eta / allocation_rule	7 / 24 / FCFS	Relationship (M1)	prt.relationship.amend
trading.availability_mode	INDICATIVE	Entity (seller)	ord.order.accept
trading.backorder_review_days	30	Entity (seller)	ord.order.accept
trading.grn_reversal_hours	24	Entity	shop.grn.reverse
trading.claim_window_days	14	Relationship	prt.relationship.amend
trading.escalation_grace_days (after window before auto-escalate)	3	Federation	fed.arbitrate
trading.invoice_consolidation	PER_GRN	Entity (seller)	bil.invoice.issue
trading.exposure_warn_thresholds	[80, 100]	Entity (seller)	bil.payment.record
trading.statement_frequency	MONTHLY	Relationship	prt.relationship.amend
trading.interest_rule (rate, grace days; manual application)	none	Relationship	prt.relationship.amend
trading.tier_basis	ORDERED_QTY (DR-2)	Federation	–

8. Screens as paper

Screen	Paper mirrored	Surface	Roles	Notes
Requisition book	Requisition book	W	Buyer stores	Suggested lines (P2) and manual; availability column; submit; amend until lock; exposure banner
Order desk	Order register	W	Seller sales	Allocation run with overrides and reasons; ETA; backorders; lock override
Delivery note	Delivery note	W	Seller dispatch	Drops per vehicle; batches from pick list; print; driver assignment
Drop list and POD	Signed DO	D	Driver	Stops in sequence; signature, photo, timestamp; offline
Goods received	GRN book	T (shop mode) / W	Receiver	Expected lines; actuals; batch data entry (MRP, expiry) with ItemCard; damaged qty; confirm; offline
Discrepancy	Discrepancy sheet	W	Both parties, arbiter	Timeline, evidence, propose/accept, escalation state
Claim	Claim form	W	Both parties, arbiter	Photos (AttachmentCapture), inspection findings, decision
Tax invoice / credit note / debit note	Tax invoice	W	Seller accounts	Read-only after issue; links; A4 print; e-invoice status
Receipt book and matching	Receipt book	W	Seller accounts	Select invoices, amounts; cheque details; reversal with reason (MFA)
Statement of account	Statement	W	Both	Period, ageing, freshness
Exposure view	Ledger summary	W	Seller accounts	Components, limit, warning state, freshness
Arbitration queue	–	W	Federation arbiter	Escalated items with both statements

9. Non-functional design

Aspect	Design
9.1 Capacity	Orders and DNs in the thousands per day; GRNs in the tens of thousands (shop receipts dominate); invoices and receipts in the thousands. All on the partitioned document base; extension tables partitioned alike. Exposure recomputation is incremental per event, not a full sum.
9.2 Offline	GRN capture and confirmation, POD capture: fully local; ingested as bundles in sequence; the expected lines for a drop travel to the shop in the snapshot on DN issue (urgent flag when the ETA is within a day).
9.3 Visibility	PARTY reads through masked views: the buyer never sees the seller's cost or margin columns, the seller never sees the buyer's retail data; discrepancy and claim evidence is visible to both; the arbiter sees both (FEDERATION_VIEW plus the arbiter role's write on arbitration only).
9.4 Immutability and links	Every issued document is immutable; every correction is a link (REVERSES, CREDITS, DEBITS, SETTLES, DISPUTES); the open balance of an invoice is derived from links, never stored except as a cached column recomputed by the same events.
9.5 E-invoicing readiness	The invoice snapshot (header, lines, both VAT numbers, tax point, hash) is what an IRD adapter will submit; e_invoice_status is metadata; nothing on the invoice changes when submission arrives (J-04).
9.6 Auditability	Every transition in Section 4 is an audit event; arbitration and reversals require MFA; the allocation run stores the availability snapshot it decided on, so a fairness dispute can be replayed (C-03).

10. Test strategy

Layer	What is tested
Unit	Every guard in Section 4; allocation FCFS and override arithmetic; lock_at computation; tier pricing by ordered quantity; open-balance derivation; exposure formula; cheque bounce reversal
Contract	Every command and query; PARTY masking (a buyer's read of an invoice has no cost columns); driver API POD offline replay
State machine	All seven machines transition by transition, including the two-step discrepancy resolution and the escalation clock
Property-based	Sum of SETTLES and CREDITS on an invoice never exceeds its total; exposure equals the sum of components under any event order; GRN received quantities are independent of DN quantities (no forced match)
Scenario	Doc 13 scenarios 1 and 2 end to end, including the offline GRN through the till simulator and ingestion; flows 6.3–6.8
Integration with M2/M5	GRN confirmation registers batches and creates lots in one committed sequence; reversal reverts lots; claim approval posts a return
Performance	Allocation run over 500 open orders under 10 s; exposure recomputation under 50 ms per event; statement generation for 800 relationships under 15 minutes

11. Decision requests

#	Point	Proposed	Decider
DR-1	Invoice timing	Follows GRN in phase 1 (per GRN or consolidated per DN); invoice-on-dispatch deferred (doc 20 DR-2)	Federation admins
DR-2	Tier basis for volume pricing	Ordered quantity (the buyer's commitment), not received	Federation with distributor input
DR-3	GRN reversal window	24 hours; no consumed lots; no invoice	Federation admins
DR-4	Escalation grace after the discrepancy window	3 days then automatic	Federation
DR-5	Backorder review age	30 days REVIEW; never auto-cancel	Federation with MPCs input
DR-6	Claim window	14 days from GRN confirmation	Federation
DR-7	Direct drop GRN without snapshot data	Capture as local supply keyed against the DN number; reconcile at ingestion	M4 and M6 designers
DR-8	Extension tables (CR-24-1 to doc 18)	Adopt as listed in Section 3	Architecture review
DR-9	Numbering rules (24B)	Assume no FY reset, GRN non-fiscal until the tax advisor answers	Tax advisor

12. Traceability

Section	Identifiers
3.1	FR-ORD-010 to 070; C-01 to C-09, C-12, C-13; A-06; ADR-04
3.2–3.3	FR-DEL-010 to 060; D-01 to D-05, D-09; A-03, A-08; ADR-03; H-02; doc 18 §5.5
3.4–3.5	D-05 to D-08; F-03; doc 18 document_link DISPUTES/CREDITS
3.6–3.7	FR-BIL-010, 030, 070, 110 to 130, 150; J-03, J-04, J-07; I-05, I-07; ADR-01
3.8–3.9	FR-BIL-020, 040, 080, 140; I-06; A-10, A-12; ADR-12; J-02
3.10	FR-BIL-050, 060; ADR-11; 24B
4	Doc 03 §2–8; C-10, C-11; D-06; H-02/H-03
6	Doc 13 scenarios 1 and 2
9	M-05 (PARTY masking); NFR-OFF, NFR-AUD

End of document. Draft for design review; the contract freezes when M5 and M6 acknowledge it, targeted end of Stage 2 week 5. Implementation guide: 24A; numbering addendum: 24B.